👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.5

# Explore Integers and Their Opposites

Lesson 7-1 · Teacher Copy

**Name:** _____ **Date:** _____ **Class:** _____

## My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can use integers to represent quantities in everyday life, explain what 0 means in each situation, and name the opposite of an integer on the number line.**

**Language Objective: I can explain my reasoning using the words integer, opposite, negative, positive, and number line.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Integer · Opposite · Negative number · Positive number · Number line · Opposite of the opposite

1. A counting number, its opposite, or 0 is an _____.

2. A number the same distance from 0 but on the other side is the _____.

3. A number less than 0, written with a – sign, is a _____.

4. A number greater than 0 is a _____.

5. A line with numbers placed in order is a _____.

6. Because –(–6) = 6, the _____ of a number is the number itself.

## Write About the Math

**Write each play as an integer, then decide: did the team make a first down, and how far are they from the line of scrimmage?**
*Escribe cada jugada como un entero y luego decide: ¿logró el equipo un primer intento, y a qué distancia están de la línea de golpeo?*

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Integer** — *Número entero*
- ☐ **Opposite** — *Opuesto*
- ☐ **Negative number** — *Número negativo*
- ☐ **Positive number** — *Número positivo*
- ☐ **Number line** — *Recta numérica*

**Model:** *In this situation 0 means ___ because ___.* — Student explains 0 is a chosen reference point (sea level), so values above it are positive and values below it are negative.

## Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The 10-yard loss is the integer ___ and the 3-yard loss is the integer ___. Combining all four plays leaves the team ___ yards from where they started, which is ___ the line of scrimmage. Because they needed at least 10 yards, the team ___ make a first down.**
- **My answer is ___ because ___.**
  *Mi respuesta es ___ porque ___.*

## Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ___.**
  *Mi afirmación es ___.*
- **My evidence is ___, so ___.**
  *Mi evidencia es ___, entonces ___.*

**Explain**

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

## Answer Key & Teacher Guide

1. **Notes 1:** Integer
2. **Notes 2:** Opposite
3. **Notes 3:** Negative number
4. **Notes 4:** Positive number
5. **Notes 5:** Number line
6. **Notes 6:** Opposite of the opposite

7. **Watch for this mistake:** *The most common mistake in this lesson is thinking that the opposite of a negative integer is still negative — for example, saying the opposite of -7 is -7, or that the opposite of -12 is -24. Opposites are the same DISTANCE from 0 but on OPPOSITE sides of 0, so taking an opposite always changes the direction and never changes the distance: the opposite of -7 is 7 and the opposite of -12 is 12. The one exception is 0, which is 0 units from itself and is therefore its own opposite. Before you submit, count the units from 0 and check that your answer is on the other side.*
8. **Try It 1:** C. 12°C and -12°C — *The opposite of -12 is 12, so 2 p.m. was 12°C. The opposite of 12 is -12, so 11 p.m. was back to -12°C.*
9. **Try It 2:** B. 12 yards past the line of scrimmage — *Write each play as an integer: -10, 4, -3, 21. Combining them gives -10 + 4 - 3 + 21 = 12, so the team ends 12 yards past the line of scrimmage.*